OpenSSL tool

Aim: To explore the OpenSSL command-line tool and study its use in encryption, decryption, hashing, key generation, and SSL/TLS connection testing.

OpenSSL is an open-source cryptographic toolkit used to perform cryptographic operations and work with SSL/TLS security protocols. It provides commands for encrypting and decrypting data, generating keys, calculating message digests, managing digital certificates, and testing secure connections.

SSL Commands

Command	Purpose
openssl version	Displays the installed version.
openssl help	Displays available commands.
openssl enc	Performs encryption and decryption.
openssl dgst	Calculates message digests or hashes.
openssl rand	Generates random bytes.
openssl req	Creates certificate requests and can generate self-signed certificates.
openssl x509	Displays and manages X.509 certificates.
openssl s_client	Tests SSL/TLS connections to servers.

Experiment 1: Check the OpenSSL version

```
┌──(praneetha㉿Praneetha)-[~/Desktop]
└─$ openssl version
OpenSSL 3.6.3 9 Jun 2026 (Library: OpenSSL 3.6.3 9 Jun 2026)
```

Result: The installed OpenSSL version is displayed.

Experiment 2: Encrypt and decrypt a text file

```
┌──(praneetha㉿Praneetha)-[~/Desktop]
└─$ echo "Hello World" > message.txt

┌──(praneetha㉿Praneetha)-[~/Desktop]
└─$ openssl enc -aes-256-cbc -salt -pbkdf2 -iter 100000 -in message.txt -out encrypted.bin
enter AES-256-CBC encryption password:
Verifying - enter AES-256-CBC encryption password:

┌──(praneetha㉿Praneetha)-[~/Desktop]
└─$ openssl enc -aes-256-cbc -d -pbkdf2 -iter 100000 -in encrypted.bin -out decrypted.txt
enter AES-256-CBC decryption password:

┌──(praneetha㉿Praneetha)-[~/Desktop]
└─$ cat decrypted.txt
Hello World
```

Result: The file was encrypted and successfully decrypted using a password-based encryption command. The same password and compatible options are required for decryption.

Experiment 3: Generate a SHA-256 hash

```
┌──(praneetha㉿Praneetha)-[~/Desktop]
└─$ openssl dgst -sha256 message.txt
SHA2-256(message.txt)= d2a84f4b8b650937ec8f73cd8be2c74add5a911ba64df27458ed8229da804a26
```

Result: OpenSSL generates a SHA-256 message digest. A hash is useful for checking file integrity, but it is not encryption and cannot be reversed to recover the original message.

Experiment 4: Generate random bytes

```
┌──(praneetha㉿Praneetha)-[~/Desktop]
└─$ openssl rand -hex 16
618537b2dda8f86066ce5434b6a3f01d
```

Result: OpenSSL generates 16 random bytes and displays them as 32 hexadecimal characters.

Experiment 5: Generate a Private Key

```
┌──(praneetha㉿Praneetha)-[~/Desktop]
└─$ openssl genrsa -out private_key.pem 2048
```

Result: An RSA private key is generated successfully and stored in the specified file.

Applications of OpenSSL

1. Encrypting and decrypting files.
2. Generating cryptographic keys and random data.
3. Calculating hashes for integrity verification.
4. Creating and examining digital certificates.
5. Testing HTTPS and SSL/TLS connections.
6. Supporting secure communication between systems.

Advantages

- Open-source and widely available.
- Supports many cryptographic algorithms.
- Provides command-line tools for common security tasks.
- Useful for learning and testing cryptographic operations.

Conclusion

OpenSSL is a useful cryptographic toolkit for performing encryption, decryption, hashing, certificate management, and SSL/TLS testing. These experiments demonstrate how it can be used to protect data and examine secure network connections